Reg. No. | | | | | | | | | | | [18CS5CT01]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

ETHICAL HACKING

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	Suppose an organization hired you for testing their security level, as an ethical hacker elaborate on all steps how do you conduct entire ethical hacking methodology for an organization?	**08**
	b.	First stage of Hacking says information gathering, how does this step help to an attacker? Discuss the objectives of foot printing.	**06**
2.	**a.**	Briefly describe about key information security concepts.	**06**
	b.	With help of different TCP/IP communication flags explain how to scan a network for open ports and service running on the machines.	**08**
3.	**a.**	TCP connection establishment is a 3-way handshaking. Justify with an example.	**08**
	b.	What is enumeration? Explain in detail about NetBIOS and SNMP enumeration.	**06**
4.	**a.**	With proper steps, describe how does kali tools and metasploit framework helps in cracking passwords? How do you generate the passwords which are difficult to crack?	**08**
	b.	There are many security policies to be implemented for each aspect of the security, what are they? Also give the steps to create and implement security policies.	**06**
5.	**a.**	Attackers usually target software vulnerabilities and configuration errors to compromise web servers. As a security professional what could be your solution to handle different kind of web server attacks.	**06**
	b.	With neat sketch elaborate on different phases of penetration testing.	**08**
6.	**a.**	You are having problems while retrieving results after performing port scanning, during internal examine. If you verify that there are no security devices between you and the target system, when both stealth and connect scanning do not work, you decide to perform a NULL scan with NMAP. The first few systems scanned shows all ports open. What is your thought on the scenario and justify your answer.	**04**
	b.	In May 2020, a New Yorker was charged for hacking into e-commerce websites	**10**

with the motive to steal credit card information. It was reported that the hacker along with its gang used SQL injection techniques for hacking into vulnerable e-commerce websites to steal payment card data. Brief about different kind of injection flaws and explain in detail how the attack was done.

7.	**a.**	An ethical hacker can only help the organization to better understand their security system, but it's up to the organization to place the right guards on the network. Justify answer.	**10**
	b.	Discuss the IT ACT-2008 sections 43, 65 and 66, explain for what reasons a person could be charged under these sections in India.	**04**
8.	**a.**	Now a day's hackers hack a system and ask ransom in the form of Bitcoins. Suppose that your website has been recently attacked by an evil hacker and there have been successful attempts of data theft. If this was the situation then to whom you will be approaching? Consider the above same scenario, if you are working professional Hacker then what could be your way of protecting the data.	**10**
	b.	"Developing Relationship" is one of the social engineering attacking phase, how important is this step? Explain briefly with example.	**04**